WEEK 1

Week 1: KVM Cluster Setup (Alternative)

HPC & Cluster Computing

Objective: Build a 3-node HPC cluster sandbox natively on Linux using KVM/QEMU and virt-manager.

Goal: Install the virtualization stack, configure an isolated virtual network, install the master node, and clone it to create compute nodes. This is an alternative to the Vagrant lab.

1. Prerequisites & Installation

This lab is designed for students running a native Linux host (e.g., Ubuntu, Fedora, Debian). Kernel-based Virtual Machine (KVM) is an open-source virtualization technology built directly into Linux.

Install KVM and virt-manager (Ubuntu/Debian)

```
sudo apt update
sudo apt install qemu-kvm libvirt-daemon-system libvirt-c
```

Note: You may need to log out and log back in for the group changes to take effect.

Install KVM and virt-manager (Fedora/RHEL)

```
sudo dnf install @virtualization
sudo systemctl enable --now libvirtd
sudo usermod -aG libvirt $USER
```

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2. Virtual Network Setup

Our cluster needs an isolated internal network to communicate without external interference.

1. Open **Virtual Machine Manager**.
2. Go to **Edit** > **Connection Details** > **Virtual Networks**
3. Click the **+** button to add a new network.
4. **Name:** `hpc-internal`.
5. **IPv4 configuration:** Network: `10.0.0.0/24`. Disable DHCP (we will use static IPs).
6. Finish creating the network.

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3. Creating the Master Node

1. Download the openEuler 24.03 ISO.
2. In `virt-manager` click **Create a new virtual machine**
3. Choose **Local install media (ISO image)** and browse to your downloaded openEuler ISO.
4. **Memory:** 2048 MB (2 GB). **CPU:** 2 Cores.
5. **Storage:** 20 GB.
6. **Name:** `hpc-master` . Before clicking Finish, check **Customize configuration before install**
7. In the hardware details, you will have one NIC (Network Interface Card) attached to the default NAT network (for internet).
8. Click **Add Hardware** > **Network**. Select the `hpc-internal` network you created.
9. Click **Begin Installation**. Install openEuler, set a root password, and create a user.

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4. Configuring the Master Node

Boot into your new `hpc-master` VM and log in.

```
# Set the hostname
sudo hostnamectl set-hostname master

# Configure the internal network interface (assuming enp
sudo nmcli connection add type ethernet con-name eth-inte
sudo nmcli connection up eth-internal
```

```
# Shut down the VM to prepare for cloning
sudo poweroff
```

5. Cloning Compute Nodes

1. In `virt-manager`, right-click the powered-off `hpc-master` VM and select **Clone**.
2. **Name:** `compute-01`. Ensure you select to clone the disk.
3. Repeat the process to create `compute-02`.

6. Reconfiguring Compute Nodes

Start `compute-01` and `compute-02`. Log into each through the virt-manager console and update their identities.

On compute-01:

```
sudo hostnamectl set-hostname compute-01
sudo nmcli connection modify eth-internal ipv4.addresses
sudo nmcli connection up eth-internal
```

On compute-02:

```
sudo hostnamectl set-hostname compute-02
sudo nmcli connection modify eth-internal ipv4.addresses
sudo nmcli connection up eth-internal
```

7. Verification & Submission

Start your Master Node. Verify cluster connectivity:

```
ping -c 3 10.0.0.101
ping -c 3 10.0.0.102
```

Submit a screenshot showing the `virt-manager` main window with all 3 VMs running, alongside your master node terminal successfully pinging the compute nodes. Save this screenshot as `Week1_Lab_Surname.png`.